Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.98314	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	79.48704	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1427.8715	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.070678	0.070678	7487.22724	0.592141	14.366667

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.98314.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 79.48704.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1427.8715.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.066394.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070678.
- Held-out runtime ms: 7487.22724.
- Robustness dispersion across held-out structure families: 0.003985.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.098281, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.094811, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.059305, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.056107, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.044885, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver-Evolution Suite

Metric	Solver Evolution	Baseline Clarke-Wright Savings	Baseline Nearest Neighbor	Baseline Regret Insertion
Feasibility Rate	1.0	1.0	1.0	1.0
Heldout Feasible Gap	**0.0707**	0.0738	0.2734	0.4664
Heldout Penalized Gap	0.0707	0.0738	0.2734	0.4664
Heldout Runtime (ms)	7487	79.5	43	1428
Robustness Dispersion	0.004	0.008	0.003	0.087
Mean Code Novelty	0.592	0.0	0.0	0.0
Mean Complexity	14.37	0.0	0.0	0.0

Analysis

- **Feasibility:** All solvers, including the evolved one, achieve perfect (100%) feasibility on held-out structure families.
- **Objective Gap:** The evolved solver achieves the lowest heldout feasible gap (0.0707), slightly better than Clarke-Wright (0.0738), and much better than Nearest Neighbor and Regret Insertion.
- **Runtime:** The evolved solver is considerably slower (~7.5 seconds) than all baselines, with Clarke-Wright being under 0.1 seconds and Nearest Neighbor under 0.05 seconds.
- **Robustness:** The evolved solver shows low robustness dispersion (0.004), comparable to baselines.
- **Novelty & Complexity:** Evolved solver introduces substantial code novelty and higher complexity compared to fixed baselines.

Conclusion

The evolved solver **consistently beats fixed baselines in objective quality on held-out problem families**, maintaining full feasibility and comparable robustness. However, this improvement comes at a significant cost in runtime and algorithmic complexity. Thus, while **the evolved solver clearly outperforms in solution quality, it does not provide a practical speed advantage over the simpler baselines.**